MILITARY SPECIFICATION

CONTROL, AUXILIARY HOIST

1. SCOPE

1.1 This specification covers auxiliary crane and hoist control units for direct and remote control, precision raising and lowering equipment.

1.2 Classification.-The control units covered by this specification shall have the following load capacities:

Size 1 - 1 Ton
Size 2 - 5 Ton
Size 3 - 10 Ton
Size 4 - 20 Ton

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-585	Boxes, Wood, Wirebound
PPP-B-591	Boxes, Fiberboard, Wood-cleated
PPP-B-621	Boxes, Wood, Nailed and Lock Corner
PPP-B-601	Boxes, Wood, Cleated Plywood

Military

MIL-P-116	Preservation, Methods of
MIL-H-5606	Hydraulic Fluid, Petroleum
MIL-T-9107	Test Reports, Preparation of
MIL-B-10377	Box, Wood, Cleated, Veneer, Paper-Overlaid
MIL-L-10547	Liners, Case, Waterproof
MIL-D-70327	Drawings, Engineering and Associated Lists

STANDARDS

Military

FSC 4920

AF - THE REPUBLICAN PRESS, Hamilton, Ohio
March 2, 1962 1500

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:13Z
Check the source to verify that this is the current version before use.

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Specifications and Standards, Order of Precedence for the Selection of
MS 28889	Valve, Air, High Pressure Charging
MS 33586	Definition of Dissimilar Metals

PUBLICATIONS

National Bureau of Standards

H 28	Screw-Thread Standards for Federal Services

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Preproduction.- This specification makes provisions for preproduction testing(see 4.5).

3.2 Components.- The control unit shall consist essentially of a reservoir, hydraulic pump, actuating cylinder, accumulator, gauges, upper and lower connection eyes, remote control apparatus, relief valve, and storage case.

3.2.1 Reservoir. - The reservoir shall have a minimum hydraulic fluid capacity of one half gallon.

3.2.2 Hydraulic Pump.- The hydraulic pump shall be manually operated, a high pressure,, single acting type, suitable for the rated and test capacity of the unit.

3.2.3 Actuating Cylinder.- The actuating cylinder with extendable piston, shall be of high grade steel and free from welds and seams. Sliding friction surfaces shall be hard surface plated and polished as required to assure long life and smooth leakage-free operation.

3.2.3.1 Vent Valve.- A breather port incorporating a dust filter shall be provided in the top of the actuator to vent the portion of the actuating cylinder above the piston to atmosphere.

3.2.4 Accumulator. - A hydraulic accumulator shall be provided

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:13Z
Check the source to verify that this is the current version before use.

which shall function as a return air "spring" for the actuator piston. Means shall be provided to prevent entry of the hydraulic fluid into the accumulator, regardless of the position of the unit.

3.2.4.1 Accumulator Valve.- The accumulator shall be provided with an air charging and relief valve and hose connector conforming to MS 28889.

3.2.5 Guages.- Two gauges shall be provided on each control unit. The gauges shall be Bourdon tube type, calibrated to read in pounds. They shall be able to withstand 25 percent overload of control unit capacity without damage to components or accuracy. The gauges shall be accurate to plus or minus one percent of full scale.

3.2.5.1 Accumulator Gauge.- The accumulator gauge shall indicate the force exerted on the actuator piston by the accumulator pressure. The gauge shall be calibrated to read in 100 pound increments.

3.2.5.2 Actuator Gauge.- The actuator gauge shall indicate the force on the piston of the actuator due to the suspended load, and shall be calibrated in the following increments:

Size 1 - 10 Pound Increments
Size 2 - 50 Pound Increments
Size 3 - 100 Pound Increments
Size 4 - 200 Pound Increments

3.2.6 Upper and Lower Connection Eyes.- The connection eyes shall be of high grade steel, machined to a good commercial finish.

3.2.7 Remote Control Apparatus.- Remote control apparatus shall be provided and shall consist of lanyards with means of attaching to the pump and relief handles on one end, and a storage reel-type handgrip on the other. The length of the lanyard shall be adjustable, and shall be at least 30 feet long when fully unreeled from the handle.

3.2.8 Relief Valve.- A manual relief valve shall be provided to relieve the pressure in the actuator cylinder. The relief valve shall closely control the flow of fluid. The valve shall throttle the flow such that the rate of piston travel shall depend upon and be approximately proportional to the valve setting. Maximum piston travel with the valve full open and the unit fully loaded shall be approximately 10 feet per minute.

3.2.9 Handles.- The pump and relief valve shall have handles spring loaded to the up and closed positions. The handles shall be suitable for convenient hand operation and shall incorporate provisions for the attachment of the remote control lanyards.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:13Z
Check the source to verify that this is the current version before use.

3.2.10 Storage Case.- A storage case shall be provided for containing and transporting the complete control unit. The storage case shall be constructed of metal or hardwood and shall be outfitted with corrosion-resistant hardware.

3.2.10.1 Castered Stands.- Castered stands suitable for accomodating the storage case with units, shall be provided with control units of sizes 3 and 4.

3.3 Selection of Specifications and Standards.- Specifications and standards for necessary commodities and services not specified herein shall be selected according to MIL-STD-143 except as provided in 3.3.1 and 3.3.2.

3.3.1 Commercial Parts.- Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, like screws, bolts, nuts, and cotter pins having suitable properties may be used provided:

a. They can be replaced by standard parts (MS and AN) without alteration.
b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the drawings.

3.3.2 Standard Parts.- With the exception of 3.3.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.4 Material.- Materials used in construction shall be of such quality as to insure conformance of the control units to requirements specified herein.

3.5 Design.- The control units shall be hydraulically operated and entirely self contained, requiring no external connections to hydraulic, electric, or air supplies during operation. The control units shall be suitable for use in any plane, horizontal through vertical. The maximum hydraulic working pressure of the unit shall not exceed 1800psi.

3.5.1 Dissimilar Metals.- Unless suitable protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MS 33586.

3.5.2 Corrosion Resistance.- Materials selected shall be corrosion resistant or shall be suitably protected by plating, painting, or other surface treatment to resist corrosion.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:13Z
Check the source to verify that this is the current version before use.

3.5.3 Threads.- All threaded parts shall comply with Handbook 28.

3.6 Interchangeability.- All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of MIL-D-70327 shall govern manufacturer's part numbers.

3.7 Construction.- The control unit shall be of safe construction, and shall have a minmum structural safety factor of 5 in the extended position. Pressure vessel minimum safety factor be 3. The construction shall be free of any characteristics or defects that may render the unit unsuitable or unsafe for the intended use.

3.7.1 Configuration.- The configuration of the unit shall be similar to Figure 1, although Figure 1 is for illustrative purposes only and is not intended to be restrictive (see Figure 1).

3.8 Performance.- The control units shall be suitable for operation by one man and have the following performance characteristics.

3.8.1 Positioning.- The control units shall be capable of raising and lowering its specified capacity through a 12 inch range.

3.8.2 Piston Travel.- The actuator piston shall have a travel between .002 and .005 inches per full pump stroke. The travel shall be easily controllable to .001 inches with partial strokes of the pump.

3.9 Seals.- All static and moving seals shall be of the "O Ring" type and suitable for use with hydraulic oil conforming to MIL-H-5606.

3.10 Safety Devices.- Each control unit shall be provided with safety devices. Parts which are hazardous to the operator, such as high pressure lines and fittings, shall be well guarded.

3.11 Physical Properties.- The control units shall conform to the following:

Size	Maximum Length (Closed Pos)	Capacity (Tons)	Maximum Weight
1	28 inches	1	125 lb
2	28 inches	5	125 lb
3	35 inches	10	320 lb
4	44 inches	20	720 lb

3.12 Marking.- The pump and relief valve handles shall be suitably marked indicating control function. The gauges shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:13Z
Check the source to verify that this is the current version before use.

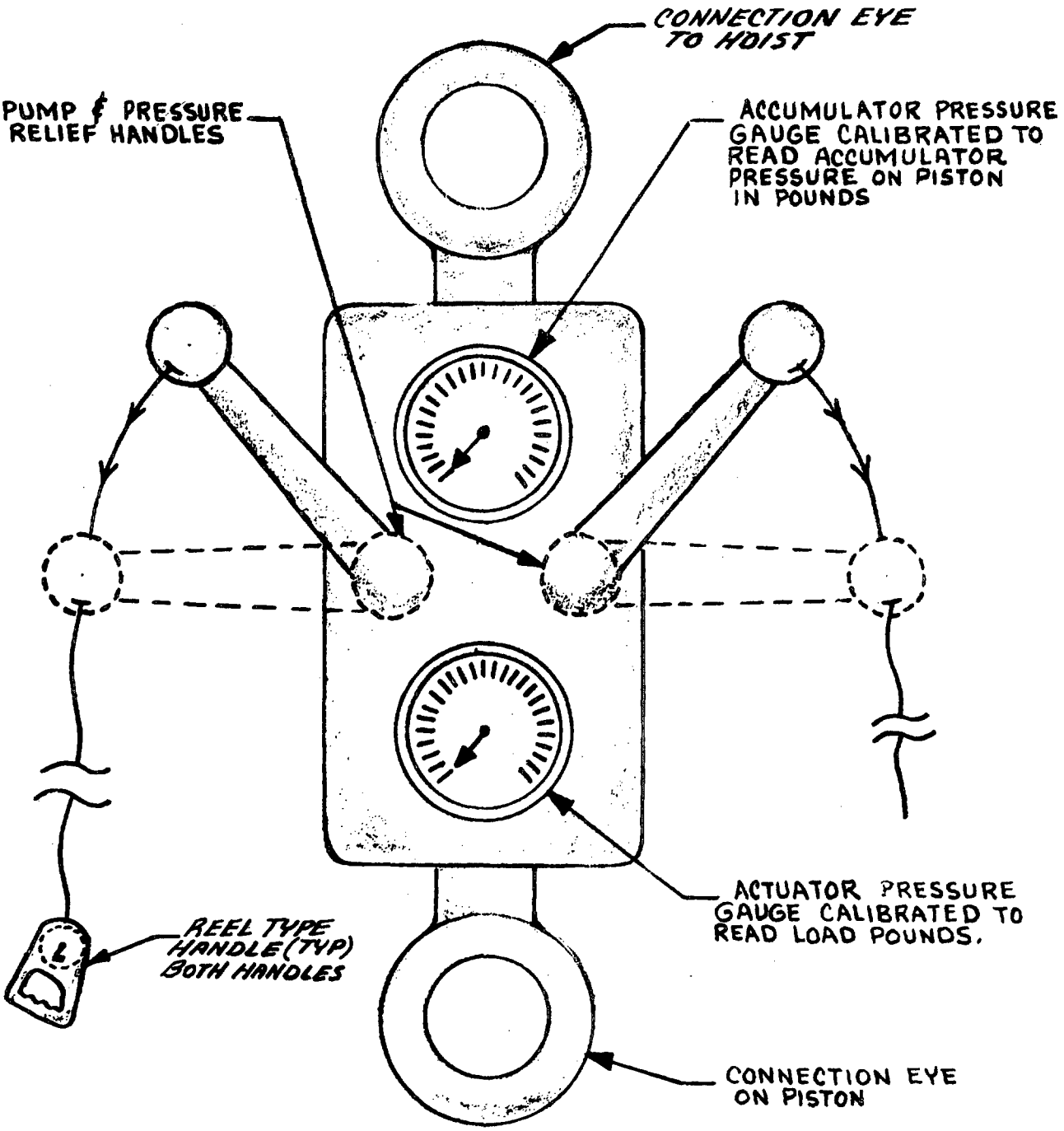


FIGURE 1

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:13Z
Check the source to verify that this is the current version before use.

marked appropriately, such as "accumulator prelowd" and "load weight".

3.13 Identification of Product.- Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.14 Painting.- Unless otherwise specified in contract or order, (see 6.2) surfaces of the control unit not otherwise finished or plated, and for which a painted surface will be suitable, shall be finished and painted in accordance with good commercial standards. Government specifications for production processes shall be used whenever they are applicable.

3.15 Workmanship.- All workmanship shall be first class to assure procurement of good quality equipment of a neat general appearance. Standards of workmanship shall be such that equipment specified herein shall comply with the requirements of Section 3. All components, subassemblies, parts, and the final assembly shall comply with the tests in Section 4.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may use his own or any other inspection facilities and services, acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Preproduction Inspection.- Preproduction inspection shall consist of all the tests specified in 4.4, and examination in accordance with 4.3.1. Preproduction inspection shall be performed by the contractor. Failure of the preproduction sample to pass the preproduction inspection shall be cause for rejection.

4.3 Acceptance Inspection.- Acceptance inspection shall consist of the examination specified in 4.3.1 and the tests specified in 4.3.2. An acceptance lot shall consist of all control units from one contractor submitted for acceptance at essentially the same time.

4.3.1 Acceptance Examination.- Each control unit submitted for acceptance shall be examined in accordance with MIL-Std-105 using the following classification of defects(see 4.3.1.1) and acceptable quality level (AQL).

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:13Z
Check the source to verify that this is the current version before use.

4.3.1.1 Classification of Defects

Major Defects

(If the control unit has one or more major defects, it shall be rejected)

101. Control unit not complete (see 3.2)
102. Evidence that the equipment will not operate (see 3.5)
103. Defective fastening devices affecting operation (see 3.4)
104. Safety devices not as specified (see 3.7)
105. Performance not as specified (see 3.8)

Minor Defects

(AQL shall be 4.0 percent defective)

201. Painting not as specified (see 3.14)
202. Marking not in accordance with MIL-STD-130 (see 3.13)
203. Packaging and packing not as specified (see Sec 5)

4.3.2 Acceptance Tests. - From the samples which have passed the acceptance examination specified in 4.3.1, samples for acceptance tests shall be selected in accordance with Inspection Level II of MIL-STD-105. Each sample shall be selected and subjected to the tests a and b below. Acceptance or rejection shall be based on an AQL of 4.0 percent defective.

a. Operational Tests (see 4.4.3)
b. Preparation for Delivery Test (see 4.4.4)

4.4 Test Methods

4.4.1 Load Test. - The control unit shall support the test load corresponding to twice its rated capacity for one hour. The unit shall be in a vertical position. The load shall be applied either by a weight or a testing machine. CAUTION: all pressure lines to the gauges shall be disconnected and plugged for this test.

4.4.2 Load Sustaining Test. - Each control unit shall sustain for 72 hours the test load corresponding to its capacity. The unit shall show no evidence of leakage or impending failure during or after the test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:13Z
Check the source to verify that this is the current version before use.

4.4.3 Operational Test. - The unit shall be cycled 25 times through its full range of travel with a capacity load attached. During the test, the units conformance with the operational requirements of paragraphs 3.2.8 and 3.8 shall be determined. At least 5 of the cycles shall be accomplished with the unit in a vertical or near vertical position. During the test the unit shall demonstrate its ability to give the operator complete and accurate control over the position of the load at all times.

4.4.4 Preparation for Delivery Test. The preservation, packaging, packing, and marking of the control unit shall be examined and tested to determine conformance with the requirements of Section 5 of this specification, and as required by specification referenced therein. Except as required by referenced specifications, inspection lots and sampling shall be in accordance with MIL-STD-105, with inspection Level L-5, and an AQL of 4.0 percent defective.

4.5 Preproduction Testing. One (1) control (see 3.1 and 6.2) shall be tested for design approval at the contractor's facility under the supervision of the procuring activity.

4.5.1 Preproduction Test Report. - The contractor shall prepare a preproduction test report in accordance with Specification MIL-T-9107 and furnish 3 copies of the report to the procuring activity.

4.5.2 Preproduction Tests. - The preproduction tests shall consist of all tests described under 4.4, conducted in the same manner.

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging. - Preservation and packaging shall be level A or C as specified by the procuring activity.

5.1.1 Level A

5.1.1.1 Cleaning. - The control unit shall be thoroughly cleaned of any residue that may cause corrosion on any part of the item. Cleaning fluids used in the cleaning process shall be removed and the item shall be thoroughly dried with dry compressed air or a clean lint-free cloth.

5.1.1.2 Packaging. - The control unit shall be packaged in its carrying and storage case with sufficient cushioning and wrapping materials to protect all glass parts, control levers, etc. The item shall be packaged in accordance with Method II, specification MIL-P-116 using the case for the immediate container.

5.1.2 Level C

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:13Z
Check the source to verify that this is the current version before use.

5.1.2.1 Cleaning. - The control unit shall be cleaned and dried in accordance with 5.1.1.

5.1.2.2 Packaging. - The control unit shall be packaged within its carrying and storage case in accordance with the manufacturer's commercial practice, except that all wrapping and packaging materials shall not cause rust or corrosion of the item.

5.2 Packing. - Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A. - Controls packaged to meet 5.1.1 shall be packed one each in exterior type shipping containers meeting PPP-B-585, PPP-B-591, PPP-B-621, PPP-B-601, or MIL-B-10377. As far as practicable, exterior containers shall be of uniform shape and size, be of minimum cube and tare consistent with the protection required. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto. Containers shall be provided with case liners conforming to MIL-L-10547, which will not be required, when due to the method of preservation, a water-vaporproof, waterproof, waterproof barrier or water resistant box will be next to the inside of the exterior container.

5.2.2 Level B. - Controls packaged to meet 5.1.1 shall be packed one each in domestic type exterior containers meeting PPP-B-601, PPP-B-585, PPP-B-591, PPP-B-621, or MIL-B-10377. Exterior containers shall be of minimum cube and tare consistent with the protection required. As far as practicable, exterior containers shall be uniform in shape and size. Containers shall be closed and strapped in accordance with the applicable container specification or appendix thereto.

5.2.3 Level C. - Packages which require overpacking for acceptance by the carrier, shall be packed in exterior type shipping containers in a manner that will insure safe transportation at the lowest rate to the point of delivery. Containers shall meet Consolidated Freight Classification rules or regulations of other common carriers as applicable to the mode of transportation.

5.3 Marking. - In addition to any marking specified in the contract or order, all interior and exterior containers shall be marked in accordance with standard MIL-STD-129.

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:13Z
Check the source to verify that this is the current version before use.

6.1 Intended Use. - The control units covered by this specification are intended for use as a link between a crane or hoist line and a load. The control unit shall be used for direct or remote control, precision raising and lowering of equipment.

6.2 Ordering Data. - Procurement documents should include the following:

a. Title, number, and date of this specification.
b. Size required (see 1.2).
c. Painting and finishing of control unit and storage case (see 3.14)
d. Level of packing and packaging required (see Section 5)

NOTICE. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any other way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any right or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing Activity SAAMA

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:13Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE C-8279

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:13Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

Commander
San Antonio Air Materiel Area
Attn: SANSS
Kelly AFB, Texas 78241

FOLD

PLATE NO. 15419 (BACK)

Source: https://assist.dla.mil -- Downloaded: 2026-10-17T13:13Z
Check the source to verify that this is the current version before use.